GAYATHRI KORAMONI

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EDUCATION

University of Southern California, Los Angeles, USA *2026 - 2028*
Master of Science in Computer Science, Viterbi School of Engineering
Relevant Coursework: *Analysis of Algorithms, Database Systems, Natural Language Processing*

Indian Institute of Technology Dharwad, Dharwad, India *2020 - 2024*
Bachelor of Technology in Electrical Engineering *CGPA: 7.69/10.00*

EXPERIENCE

Software Engineer, Acies, Goa, India *July 2024 - Jan 2026*

- Built a full analytics dashboard platform from scratch (Dataflows, Charts, Apps modules) using **React**, **MUI**, and **Django**, supporting **30+** chart types across **Plotly.js**, **Apache ECharts**, and **Recharts** — including candlestick, boxplot, sankey, and sunburst visualizations for financial and operational data.
- Integrated pivot-table functionality (**react-pivot**) for data cleanup and validation, and built logic to merge and reconcile data across multiple heterogeneous sources before feeding downstream visualizations.
- Designed and built a job scheduling application in React and Bootstrap to schedule, track, and manage recurring jobs, with status tracking for successful/failed executions.
- Proposed and implemented a **React Flow**–based pipeline visualization exposing data lineage and transformations to non-technical stakeholders, improving trust and transparency in dashboard outputs.

Tech Stack: *React, MUI, Django, React Flow, Plotly.js, Echarts, Charts.js, Bootstrap*

Full stack intern, Doutya Technologies, Bangalore, India *Nov 2023 - June 2024*

- Built full-stack web and mobile applications using **React**, **Node.js**, **React Native**, and **PHP**, including admin dashboards on **Next.js** with **SQL**-backed data management.
- Integrated **Firebase** for real-time backend data synchronization across web and mobile clients.
- Developed immersive AR/VR application experiences using **Unity** and **ViroReact** for location-based visualization.

Tech Stack: *React.js, Node.js, React Native, PHP, Next.js, Firebase, Unity, ViroReact, SQL*

Research and Development, IIT Dharwad, Dharwad, India *Aug 2023 - Oct 2023*

- Benchmarked **ML (Linear Regression)** against **DL architectures** (ConvLSTM+Conv3D, 1D-CNN, Conv3D) for vegetative drought forecasting using **NOAA STAR NESDIS satellite data** and gridded precipitation data (1981–2021, Karnataka ROI).
- Found the ML model outperformed DL by up to **70 percentage points in R^2** (0.92 vs. 0.22–0.83) while cutting training time from **300 minutes to under 2 minutes.**

PROJECTS

Data Reconciliation Copilot

- Built a web app to **reconcile multi-source tabular data**, using fuzzy column matching (rapidfuzz) with LLM-confirmed mappings to flag **formatting issues, duplicates, and conflicting records.**
- Integrated an **LLM API** to generate **plain-English reconciliation reports** and answer **natural-language follow-up queries** grounded in computed diffs.

Tech Stack: *Python, FastAPI, React, TypeScript, PostgreSQL, Anthropic/OpenAI API, rapidfuzz*

Robot Digital Twin

- Built a **browser-based 3D simulation** of an autonomous grass-cutting robot, streaming **live telemetry** (battery, position, sensors) over **WebSockets** to a real-time dashboard.
- Implemented **coverage path planning** with **A*-based obstacle avoidance**, storing run history in **PostgreSQL** to compare strategies.

Tech Stack: *Three.js, React, TypeScript, Node.js, WebSockets, PostgreSQL, Docker Compose*

Topic Modelling using LDA, link

- Applied **Latent Dirichlet Allocation (LDA)** on a complaints dataset to extract keywords and uncover hidden topics, using **spaCy**, **NLTK**, and regex for preprocessing.
- Performed **sentiment analysis** on extracted topics to evaluate user opinions and identify patterns across complaint categories.

SKILLS

Languages & Skills: Python | JavaScript | TypeScript | React | ReactNative | Vue.js | Tableau | Power BI | Django | C++ | Java | Linux |
Frameworks and tools: Node.js | Docker | Kubernetes | MongoDB | MySQL | Git | Github | Heroku | Postman | Babel | RestAPIs | Azure |
AI/ML Tools: LLM APIs (OpenAI, Anthropic) | LangChain | Prompt Engineering | RAG
Areas of Interest: Web Design and Development, Frontend, Backend, Machine Learning, Data Science